CAITLIN DAMARSINGH

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Mechanical and Aerospace Engineering student with hands-on experience in rocketry, mechanical design, sensor integration, manufacturing, electronics, and engineering research and leadership.

EDUCATION

University of Florida (UF) — B.S. Mechanical Engineering & B.S. Aerospace Engineering (Dual Degree) *Expected Dec. 2028*
Gainesville, FL | GPA: 3.57

ENGINEERING & RESEARCH EXPERIENCE

Swamp Launch Rocket Team *May 2025 – Present*
University of Florida

Payload Mechanics Responsible Engineer — Sensor Suite (IREC Competition) *Sept. 2026 – Present*
- Lead mechanical design and integration of a sensor suite within a CubeSat payload that measures temperature, pressure, and vibration during flight
- Design CAD mounts and sliders that secure sensor cards in flight and deploy sensors through CubeSat slots for external sensing.

Tooling Development Responsible Engineer — Load Cell (Swamp Launch Development) *Sept. 2026 – Present*
- Own the load-cell subsystem for a solid-rocket-motor static-fire test stand, defining mounting and installation requirements for thrust measurement.
- Develop the load-cell integration and verification approach, coordinating mechanical interfaces, mounting requirements, sensor placement, and validation with the broader test-stand design and manufacturing effort.

Payload Electronics Associate Lead (NASA USLI Competition) *May 2025 – June 2026*
- Designed and integrated NASA USLI payload electronics for a high-power rocket and payload through formal safety and design reviews.
- Built custom PCBs, integrated sensors, soldered hardware, and troubleshot system-level issues during payload testing.
- Designed and 3D-printed three electronics sleds and associated mounting hardware in CAD to package, secure, and organize payload electronics while maintaining proper component placement and mechanical integration within the payload structure.
- Led soldering workshops and coordinated electrical-mechanical integration with payload mechanics and software members.

Undergraduate Researcher — Dr. Simmons Research Group, Civil & Coastal Engineering *Sept. 2026 – Present*
University of Florida
- Research "verification before reliance" in AI-enabled engineering practice, examining how engineers evaluate AI-generated information before incorporating it into technical work and decisions.
- Conduct structured literature reviews across engineering, AI/HCI, and human-factors research to identify verification behaviors, trust-calibration factors, and conditions that shape verification.
- Synthesize findings into annotated bibliographies, concept maps, research summaries, and presentation materials used to refine the team's research questions and conceptual framework.

Student Assistant — EGM 2511 (Engineering Mechanics: Statics), Mechanical & Aerospace Engineering *Sept. 2026 – Present*
University of Florida
- Grade Statics assignments and recitations using instructor rubrics, evaluating solution methodology, calculations, and overall technical quality..
- Review student work for use of engineering principles and unit consistency, while applying grading standards consistently across submissions.
- Assist students with Statics concepts and grading-related questions by clarifying solution methods, explaining problem-solving approaches, and answering questions about course material.

LEADERSHIP & ACTIVITIES

External Vice President | Former Social Chair — Indo-Caribbean Student Association *May 2025 – Present*
University of Florida
- Lead external outreach, partnerships, and communication with campus and community organizations to support programming and visibility.
- Previously planned and executed two or more Indo-Caribbean cultural and social events per month with accurate heritage representation.
- Manage event logistics from planning through closeout, including scheduling, venues, funding, vendors, volunteers, setup, and partner coordination.

Student Senator — Student Government *June 2025 – Aug. 2025*
University of Florida
- Selected through an interview process to represent on-campus undergraduates and participated in weekly Senate meetings, legislative discussion, and voting.
- Collaborated with fellow senators to review proposed legislation and contribute analysis and discussion on issues affecting the student body.
- Participated in review of an approximately $23M Student Government budget and authored, presented, and debated a budget amendment.

Lead Validator — Science CommuniGator *Aug. 2025 – Present*
University of Florida
- Review student-written science articles for factual accuracy, currency, and source quality before publication for public audiences.
- Validate references and guide writers to credible scientific sources, strengthening claims and technical accuracy while meeting deadlines.

TECHNICAL SKILLS & CERTIFICATIONS

CAD / Engineering: SOLIDWORKS (CSWA), Fusion 360, MATLAB | **Electronics:** Arduino, Altium, Raspberry Pi
Manufacturing: CNC machining, 3-axis mills, lathes, drilling, tapping, reaming, general shop fabrication and assembly
Certification: L1 High Power Rocketry Certification (In Progress)

AWARDS & EXTRA-CURRICULARS

University of Florida Herbert Wertheim College of Engineering Dean's List (Fall 2025, Spring 2026) • C. Addison Pound Scholarship Recipient for Fall 2025 — Herbert Wertheim College of Engineering • P.M. Pope Scholarship Recipient for Fall 2026 — Herbert Wertheim College of Engineering